

Bike Sharing

Provide your solution in a small report of max 4 pages

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Background

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Bike sharing systems are new generation of traditional bike rentals where whole process from membership, rental and return back has become automatic. Through these systems, user is able to easily rent a bike from a particular position and return back at another position. Currently, there are about over 500 bike-sharing programs around the world which is composed of over 500 thousand bicycles. Today, there exists great interest in these systems due to their important role in traffic, environmental and health issues.

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Data Set and Task

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Bike-sharing rental process is highly correlated to the environmental and seasonal settings. For instance, weather conditions, precipitation, day of week, season, hour of the day, etc. can affect the rental behaviors. The core data set is related to the two-year historical log corresponding to years 2011 and 2012 from Capital Bikeshare system, Washington D.C., USA which is publicly available in <http://capitalbikeshare.com/system-data>. The corresponding weather and seasonal information are extracted from <http://www.freemeteo.com>.

The general task is to predict the daily bike rental count based on the environmental and seasonal settings. A more specific task is to understand if there is any difference in the use of bike-sharing services in holiday if compared with working days.

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Variables

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- instant: record index
- dteday : date
- season : season (1:springer, 2:summer, 3:fall, 4:winter)
- yr : year (0: 2011, 1:2012)
- mnth : month (1 to 12)
- holiday : whether day is holiday (1) or not (0) (from <http://dchr.dc.gov/page/holiday-schedule>)
- weekday : day of the week
- workingday : if day is neither weekend nor holiday is 1, otherwise is 0.
- + weathersit :
 - 1: Clear, Few clouds, Partly cloudy, Partly cloudy
 - 2: Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist

- 3: Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds
- 4: Heavy Rain + Ice Pallets + Thunderstorm + Mist, Snow + Fog
- temp : Normalized temperature in Celsius. The values are divided to 41 (max)
- atemp: Normalized feeling temperature in Celsius. The values are divided to 50 (max)
- hum: Normalized humidity. The values are divided to 100 (max)
- windspeed: Normalized wind speed. The values are divided to 67 (max)
- casual: count of casual users
- registered: count of registered users
- count: count of total rental bikes including both casual and registered

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Reference publication

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[1] Fanaee-T, Hadi, and Gama, Joao, "Event labeling combining ensemble detectors and background knowledge", Progress in Artificial Intelligence (2013): pp. 1-15, Springer Berlin Heidelberg, doi:10.1007/s13748-013-0040-3.

To read the file:

```
mydata <- read.csv("C:\\Users\\User\\Documents\\Applied\\Finals\\Bike_Sharing\\BS_day.csv")
```